

Hamza Ouriour

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SUMMARY

Mechanical and Robotics Engineering student with two NASA internships and hands-on reactor, vehicle, and facility hardware experience. I design in CAD, build and integrate hardware, and test it until I understand why it works or fails.

EDUCATION

Wentworth Institute of Technology

Exp. Graduation: August 2027

B.S. Mechanical Engineering & Robotics Engineering, Minor in Aerospace Engineering | GPA: 3.75 | Presidential Scholarship Boston, MA

Relevant Coursework: Thermodynamics, Fluid Mechanics, Dynamics, Mechanics of Materials, Robotics & Controls, Numerical Methods

HARDWARE & TEST EXPERIENCE

Clock Tower Engineering

Jul 2026 – Present

Mechanical Engineering Intern

- Developing and iterating a plasma-assisted fluidized bed reactor (PFBR) that fluidizes fine metal-oxide particles and exposes them to reactive gas and plasma to speed up hydrogen-based metal-oxide reduction.
- Reworked the reactor through multiple design iterations into a more compact, integrated system with simpler external tubing and flow paths, while keeping gas flow, cooling, plasma, and particle handling intact.
- Troubleshooting thermal management, plasma stability, electrical arcing, particle agglomeration, gas-flow control, and cyclone separation, including water-cooled torch components and improved instrumentation.
- Improved overall reactor operating efficiency by about 35% through iterative redesign and system integration.

NASA Johnson Space Center

May 2025 – Sep 2025

Robotics Engineering Intern

Houston, TX

- Assembled and integrated hardware for the Lunar Terrain Vehicle Ground Test Unit (LTV GTU), a full-scale Artemis testbed for lunar surface evaluation.
- Managed part procurement, revision cycles, and component-level assembly across 6+ subsystems while keeping configuration records traceable.
- Ran testbed setup, functional verification, and maintenance with a cross-functional team to keep the vehicle ready for simulation testing.

Hallam-ICS

Dec 2025 – Jun 2026

Mechanical Engineer Intern

Boston, MA

- Developed process gas, cooling water, and exhaust system layouts for semiconductor and research facilities and supported P&ID and equipment cut-sheet reviews.
- Produced fabrication-ready layouts and construction drawings in Revit and AutoCAD.

Starship Robotics

Aug 2023 – Apr 2024

Robotics Engineer & Data Scientist

Boston, MA

- Designed and validated sensor-actuator electronics, improving closed-loop performance through iterative hardware testing.
- Analyzed operational sensor data to find failure patterns and wrote reports that guided hardware and software changes.

ENGINEERING & ANALYSIS EXPERIENCE

NASA Europa Clipper Mission

Jun 2024 – Jan 2025

EIS Optimization Engineer Intern

Hybrid

- Developed imaging optimization algorithms for the Europa Imaging System (EIS), improving frame-selection efficiency by ~23%.

Durobotics

Oct 2025 – Present

Founder & Lead Engineer

Boston, MA

- Built a physics simulation of BLDC motor and ball-screw electromechanical actuators with Weibull and Arrhenius degradation models.
- Trained Random Forest and Gradient Boosting models on 17 physics-based features, reaching 94.2% fault classification accuracy and 97% critical-fault recall.

PROJECTS

Autonomous Precision UAV Landing & Navigation System | PX4 SITL, Gazebo, ROS, OpenCV, MAVSDK, Python

- Built a vision-guided landing pipeline for GPS-denied quadrotors with under 10 cm dispersion across 70+ simulation runs.
- Resolved NED/ENU frame errors in the LiDAR and vision stack to hold a stable 2 m hover under simulated crosswind.

UAV LiDAR Autonomy | ROS Noetic, Gazebo, Velodyne VLP-16, Python

- Wrote a perception and navigation stack that flies a quadrotor through a cluttered 16 m x 16 m warehouse with no GPS or prebuilt map.

TECHNICAL SKILLS

CAD & Design: SolidWorks, Siemens NX, AutoCAD, Revit, Construction Drawings, P&ID Review

Hardware & Test: Data Acquisition, Prototyping, Embedded Systems, Soldering, 3D Printing, CNC Mill, Optical Comparator, Laser Marking

Software & Analysis: Python, MATLAB, C++, NumPy, ROS, PX4, Gazebo, OpenCV, MAVSDK, SQL

Certifications: CSWA (SolidWorks), Siemens NX Certified